

Experiment No 1

Aim : Write a program to implement naive bayes classifier for a simple training database as saved as 'titanic.csv' compute the accuracy of classifiers considering few test datasets

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```
[1]: import pandas as pd
```

```
[2]: df=pd.read_csv("titanic.csv")
```

```
[3]: df
```

```
[3]:
```

	Survived	Pclass	Name	Sex	Age	Siblings/Spouses Aboard	Parents/Children Aboard	Fare
0	0	3	Mr. Owen Harris Braund	male	22.0	1	0	7.2500
1	1	1	Mrs. John Bradley (Florence Briggs Thayer) Cum...	female	38.0	1	0	71.2833
2	1	3	Miss. Laina Heikkinen	female	26.0	0	0	7.9250
3	1	1	Mrs. Jacques Heath (Lily May Peel) Futrelle	female	35.0	1	0	53.1000
4	0	3	Mr. William Henry Allen	male	35.0	0	0	8.0500
...	...	...	...	...	...	...	...	...
882	0	2	Rev. Juozas Montvila	male	27.0	0	0	13.0000
883	1	1	Miss. Margaret Edith Graham	female	19.0	0	0	30.0000
884	0	3	Miss. Catherine Helen Johnston	female	7.0	1	2	23.4500
885	1	1	Mr. Karl Howell Behr	male	26.0	0	0	30.0000
886	0	3	Mr. Patrick Dooley	male	32.0	0	0	7.7500

887 rows x 8 columns

```
[4]: df.drop(["Name", "Siblings/Spouses Aboard", "Parents/Children Aboard"], inplace=True, axis="columns")
```

```
[5]: df
```

```
[5]:
```

	Survived	Pclass	Sex	Age	Fare
0	0	3	male	22.0	7.2500
1	1	1	female	38.0	71.2833
2	1	3	female	26.0	7.9250
3	1	1	female	35.0	53.1000
4	0	3	male	35.0	8.0500
...	...	...	...	...	...
882	0	2	male	27.0	13.0000
883	1	1	female	19.0	30.0000
884	0	3	female	7.0	23.4500
885	1	1	male	26.0	30.0000
886	0	3	male	32.0	7.7500

887 rows × 5 columns

```
[6]: Target=df.Survived
```

```
[7]: Target
```

```
[7]:
```

0	0
1	1
2	1
3	1
4	0
...	...
882	0
883	1
884	0

```
885    1
886    0
Name: Survived, Length: 887, dtype: int64
```

```
[8]: Inputs=df.drop("Survived",axis="columns")
```

```
[9]: print(Inputs)
```

	Pclass	Sex	Age	Fare
0	3	male	22.0	7.2500
1	1	female	38.0	71.2833
2	3	female	26.0	7.9250
3	1	female	35.0	53.1000
4	3	male	35.0	8.0500
..	...	...	...	...
882	2	male	27.0	13.0000
883	1	female	19.0	30.0000
884	3	female	7.0	23.4500
885	1	male	26.0	30.0000
886	3	male	32.0	7.7500

[887 rows x 4 columns]

```
[10]: dummy=pd.get_dummies(Inputs.Sex,dtype=int)
```

```
[11]: dummy
```

```
[11]:
```

	female	male
0	0	1
1	1	0
2	1	0
3	1	0
4	0	1
...	...	...
882	0	1
883	1	0
884	1	0

886 0 1

887 rows × 2 columns

```
[12]: Inputs=pd.concat([Inputs,dummy],axis="columns")
```

```
[13]: Inputs
```

```
[13]:
```

	Pclass	Sex	Age	Fare	female	male
0	3	male	22.0	7.2500	0	1
1	1	female	38.0	71.2833	1	0
2	3	female	26.0	7.9250	1	0
3	1	female	35.0	53.1000	1	0
4	3	male	35.0	8.0500	0	1
...	...	...	...	...	...	...
882	2	male	27.0	13.0000	0	1
883	1	female	19.0	30.0000	1	0
884	3	female	7.0	23.4500	1	0
885	1	male	26.0	30.0000	0	1
886	3	male	32.0	7.7500	0	1

887 rows × 6 columns

```
[14]: Inputs=Inputs.drop("Sex",axis="columns")
```

```
[15]: Inputs
```

```
[15]:
```

	Pclass	Age	Fare	female	male
0	3	22.0	7.2500	0	1
1	1	38.0	71.2833	1	0

```
[15]:
```

	Pclass	Age	Fare	female	male
0	3	22.0	7.2500	0	1
1	1	38.0	71.2833	1	0
2	3	26.0	7.9250	1	0
3	1	35.0	53.1000	1	0
4	3	35.0	8.0500	0	1
...	...	...	...	...	...
882	2	27.0	13.0000	0	1
883	1	19.0	30.0000	1	0
884	3	7.0	23.4500	1	0
885	1	26.0	30.0000	0	1
886	3	32.0	7.7500	0	1

887 rows × 5 columns

```
[16]: from sklearn.naive_bayes import GaussianNB
```

```
[17]: model=GaussianNB()
```

```
[18]: model
```

```
[18]: GaussianNB
GaussianNB()
```

```
[19]: model.fit(Inputs,Target)
```

```
[19]: GaussianNB
GaussianNB()
```

[19]: GaussianNB
GaussianNB()

[48]: Inputs

[48]:

	Pclass	Age	Fare	female	male
0	3	22.0	7.2500	0	1
1	1	38.0	71.2833	1	0
2	3	26.0	7.9250	1	0
3	1	35.0	53.1000	1	0
4	3	35.0	8.0500	0	1
...	...	...	...	...	...
882	2	27.0	13.0000	0	1
883	1	19.0	30.0000	1	0
884	3	7.0	23.4500	1	0
885	1	26.0	30.0000	0	1
886	3	32.0	7.7500	0	1

887 rows × 5 columns

Viva Questions

1. What was the objective of using the Titanic dataset in this practical? -> The objective was to predict whether a passenger survived or not based on given features.
2. Which column was taken as the target variable in the Titanic dataset? -> The Survived column was taken as the target variable.
3. Name any two features used from the Titanic dataset for prediction. -> Age and Sex were used as features.
4. Why was data preprocessing required before applying Naive Bayes? -> Preprocessing was required to handle missing values and convert categorical data into numerical form.

Viva Questions

1. What was the objective of using the Titanic dataset in this practical? -> The objective was to predict whether a passenger survived or not based on given features.
2. Which column was taken as the target variable in the Titanic dataset? -> The Survived column was taken as the target variable.
3. Name any two features used from the Titanic dataset for prediction. -> Age and Sex were used as features.
4. Why was data preprocessing required before applying Naive Bayes? -> Preprocessing was required to handle missing values and convert categorical data into numerical form.
5. Which Naive Bayes model is commonly used for the Titanic dataset and why? -> Gaussian Naive Bayes is used because many features like Age and Fare are continuous.

Conclusion : In this practical, we used Naive Bayes on the Titanic dataset using Python to predict passenger survival. After preprocessing the data, the model was trained and tested successfully, showing that Naive Bayes is a simple and efficient classification algorithm.